

Virtual Digital Assistant (VDA)

AI-guided NCD care navigation in rural India

Challenge	AI-enabled Virtual Digital Assistant — AI for Ambulatory Care
Brief issued	Friday, 31 July 2026 — end of day
Submission deadline	Monday, 3 August 2026 — 12:00 Noon IST
In-person round	Monday, 3 August 2026 — 5 to 10 minutes, from 12:00 Noon IST onwards. In person in Bengaluru or over video; slot to be confirmed.

You are being asked to show us how you think, where you choose to spend your judgement, and what you refuse to compromise on when a system is going to speak to a patient who cannot read.

Build something you would be glad to demo. Then come tell us about it.

Submission ownership

Candidates retain ownership of all materials submitted. Submissions will be used solely for evaluating candidates for this hiring process and will not be incorporated into production products or commercial offerings without a separate written agreement.

Confidentiality

This brief is confidential and intended only for candidates participating in this hiring process.

Hiring

Completion of this challenge does not guarantee an interview, offer, or employment.

Expenses

Candidates are not expected to incur costs beyond freely available tools, and no reimbursement will be provided.

1 · Read this first

Use AI tools. Aggressively.

We build with AI here, so we hire people who are good at building with AI. Claude, Cursor, Copilot, Codex, whatever you like — use all of it. There is no purity test and no hidden penalty.

Cost

Stay on free tiers only. Spending will not improve your score.

2 · Why this matters

India's public health system reaches an extraordinary number of people. What it struggles to do is stay in contact with a patient between visits — and non-communicable disease care is almost entirely made of the space between visits. A hypertensive patient sees a clinician for perhaps twenty minutes a quarter and then manages the other 129,580 minutes alone.

The evidence on what happens in that gap is uncomfortable:

What the research shows	Where it hurts
Only 23.7% of hypertensive patients had regular consultations. 13% took medication only, 16.3% were irregular, and 47% had no treatment at all. The single largest cause of missed doses was simply forgetting.	Use case 1 — Adherence
48% of NCD patients did not seek care because they did not consider their illness serious.	Use case 1 — Adherence
56% of beneficiaries were unaware of the treatment available to them under PM-JAY. Of those trying to close that gap, 8% used the toll-free number, 51% needed in-person help, and 49% sought no support at all.	Use case 2 — Scheme entitlement
Only 10.8% of people who died of acute cardiac events or stroke reached an appropriate facility within one hour. Of those delayed, 40% went to the wrong facility first.	Use case 3 — Facility linkage
Nearly two-thirds of TB patients experienced diagnostic delay. Seeing more than three providers tripled the odds of delay.	Use case 3 — Facility linkage

Read those together and a pattern falls out. Very little of this is a medicine problem. It is an information-routing problem — the right guidance, in the right language, at the moment the patient actually has the question, which is rarely during a consultation and often at 9pm on a Tuesday.

What VDA is meant to change

- **A patient who cannot read** can still ask a question, out loud, in her own dialect, and get a straight answer.
- **A patient who forgot** gets a nudge before the dose is missed, not a scolding at the next visit.
- **A patient entitled to free treatment** finds out that she is entitled to it.
- **A patient with chest pain at 2am** is told to go to a facility that can actually treat her — and a human being is alerted.

The design constraint that makes this hard and interesting

Every one of those four sentences is a sentence an LLM can produce fluently and get wrong. Fluency is cheap now; being reliably right in a language you cannot read back, for a user who has no way to sanity-check you, is not. That gap is the whole engineering problem, and it is what this weekend is about.

3 · The product, in one page

VDA is a voice-first assistant embedded as a mini-app inside an existing patient application — no separate download, no typing, no reading required. It is scoped to four use cases and is deliberately forbidden from doing anything else.

	Use case	What it does
UC1	NCD care adherence	Explains the follow-up and medication schedule in plain language; daily medication reminders; 3-day follow-up alert; annual NCD screening nudge; health guidance drawn from government protocols.
UC2	Scheme entitlement check	Checks eligibility against the patient's profile, walks them through enrolment, and creates awareness of what they are entitled to.
UC3	Public health service linkage	Finds a public facility that provides the service the patient needs, near where they are.
UC4	Teleconsultation & triage	Connects to the existing teleconsultation portal for timely advice and appropriate referral; escalates red flags to a clinician.

A reference pipeline

Six steps, from a spoken sentence to a spoken answer. Treat this as one reasonable shape rather than a specification — it is here so you know roughly what the moving parts are. A handful of points are genuinely non-negotiable and are marked as such. Everything else is yours to design, and a different structure you can defend is a perfectly good answer.

#	Step	Notes
01	Patient speaks (mini-app)	Voice-first. Removes the literacy barrier — no reading, no typing, no separate app.
02	Speech to text — e.g. Google Cloud STT free tier	May cover the major Indian languages and cope with dialect and Hindi–English code-switching. Detect the language before transcribing. Engine choice is yours — keep it behind an interface so it can be swapped.
03	Intent classification	Sends the question to the right use case. Non-negotiable: low-confidence input falls back deterministically — never to a guess.
04	Agent + RAG	Non-negotiable: answers come from retrieved context, not from model memory. How you organise and isolate the index is up to you — tell us why.
05	Safety gate	Deterministic rules fire first. Non-negotiable: the LLM never decides escalation and can never override the gate. A missed escalation is the failure that matters most — set a target, measure it, and report it honestly.
06	Text to speech — e.g. Google Cloud TTS free tier	The response is translated back into the patient's language and synthesised as audio. Latency matters: this has to feel like a conversation on a mid-range Android phone over a patchy mobile connection.

What VDA must never do

- Diagnose, prescribe, or interpret lab results.
- Generate free-form clinical advice.
- Answer anything outside the four use cases.
- Retain patient PII after the session ends.
- Allow the LLM to override a deterministic escalation.